Jason Seh

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EDUCATION

B.S in Computer Science and Engineering, The Ohio State University *Expected May 2029*

RESEARCH & PUBLICATIONS

[1] J. Seh, B. Yogananda, S. Chowdhury, and H. Subramoni, "The Workflow Is the Curriculum: Provenance-Aware Learning Modules from AI/HPC Artifacts," submitted to IEEE HiPC Workshops (EduHiPC) 2026.

EXPERIENCE

AI/ML Research Engineer **Columbus, OH**

NSF ICICLE AI Institute | Python, PyTorch, React, Docker, Kubernetes, NGINX, W&B, Qdrant *Apr 2026 - Present*

- Developed a distributed GPU-accelerated AI pipeline on Kubernetes and Tapis HPC for zero-shot image classification, segmentation, and explainable AI using DINOv2, SAM, and Grad-CAM.
- Authored a first-author systems paper introducing a provenance-driven artifact contract that transforms AI/HPC workflow artifacts into traceable learning modules.
- Engineered an end-to-end RAG pipeline using vLLM to serve open-weight models on Kubernetes, deploying Qdrant on Tapis Pods with persistent storage for semantic retrieval and self-healing PyTorch code generation.
- Sustained **~91% GPU utilization** during large-scale inference by implementing multithreaded PyTorch DataLoaders with lazy loading, and reduced cross-validation training time by **~75%** across 4 GPUs via Python multiprocessing.
- Built and deployed a containerized full-stack app using Vite, React, and Nginx on Tapis Pods, featuring pipeline tailing, an interactive YAML configuration generator, and automated OAuth token lifecycle management for HPC job submission.

AI Engineering Intern **Columbus, OH**

Center for Software Innovation | AWS, Docker, Python, YOLO, LiteLLM, FastAPI, React *Sep 2026 - Present*

- Designed an automated competitive benchmarking pipeline for Honda R&D using FastAPI, LLMs, and Playwright, cutting vehicle seating data analysis time by **50%** and enabling same-day report generation.
- Automated end-to-end data collection, processing, and analysis through a multimodal AI orchestration layer using YOLO and LiteLLM to streamline competitive benchmarking and accelerate actionable insights and report generation.
- Optimized LLM context efficiency by **85.5%** by using a data sanitizer and token clustering algorithm, cutting inference costs.

Technical Program Volunteer – AI/ML **Columbus, OH**

The Ohio Supercomputer Center | Python, PyTorch, DINOv2, YOLO, Slurm, Bash, ONNX *May 2026*

- Selected by OSC leadership to develop curriculum, starter codebases, and solution repositories for the "AI in the Field" program.
- Converted computer vision models to ONNX and optimized GPU inference pipelines for OSC Ascend and Cardinal HPC clusters, reducing inference latency by **~66%**

PROJECTS

Splice - AI-Powered Binary Reconstruction Platform | ***AWS EC2, Bedrock, S3, Systems Manager, Docker, Ghidra, OpenCode***

- Built an AI-powered binary reconstruction platform to analyze compiled applications, recover behavioral specifications, and synthesize reusable software components; won **Most Original at HackOHI/O 2026**.
- Engineered an on-demand AWS pipeline using EC2, Systems Manager, and Docker to run containerized Ghidra and REA analysis, featuring live progress streaming, persistent S3 artifact storage, and automated worker startup and shutdown.
- Orchestrated Amazon Bedrock–powered agents through OpenCode and MCP to inspect decompiled code, trace function relationships, draft specifications, and independently review findings, with restricted tool access and traceable binary evidence.

AceIt - HackAI | ***Next.js, FastAPI, WebRTC, PyTorch, Librosa, Whisper, Gemini, Supabase***

- Built an AI-powered mock interview platform that used real-time computer vision and audio analysis to evaluate candidate performance across behavioral and technical interviews; placed **7th** among 100+ teams.
- Engineered an asynchronous audio pipeline using Whisper, Librosa, and PyTorch, reducing response latency by **~80%**
- Built an asynchronous backend with aiohttp and aiortc for low-latency media streaming and concurrent session processing, storing interview analytics in Supabase.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, Java, Bash
Frameworks & Libraries: FastAPI, Node.js, Next.js, React, Vite
AI/ML: PyTorch, XGBoost, DINOv2, SAM, Transformers, vLLM, RAG, Scikit-Learn, W&B, MCP
Infrastructure & Tools: Docker, Kubernetes, AWS, NGINX, Grafana, Slurm, Git, GitHub Actions, Linux, Vercel, Cloudflare R2
Database: PostgreSQL, PostGIS, Supabase, ChromaDB, Qdrant, MongoDB